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kayena vaacha manasendriyairva

buddhyatmana va prakrte swabhavath

karomi yad yad sakalam parasmai

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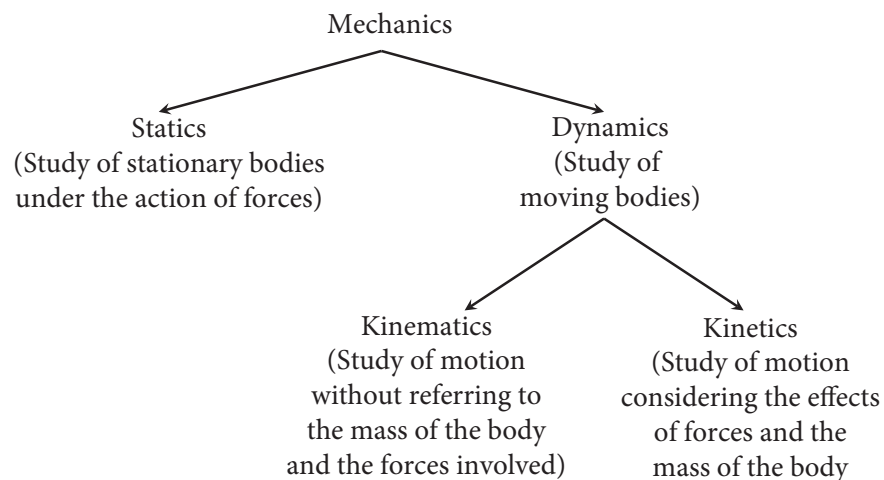
Preface

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Fundamentals of Mechanics

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Rigid Body

When the deformation of a body is assumed to be negligibly small and not significant enough to affect the results, the body is said to be **rigid**. Bodies in the real world deform a little. But if the deformation is neglected, many an engineering problem can be tackled easily and the solutions become simpler without compromising their accuracy. The relative distance between any two points in a rigid body will be the same though the body as a whole can be in motion. Such motion is called Rigid Body Motion (RBM).

Basic Laws of Mechanics

The following laws form the foundation of the science of mechanics.

Newton's First Law

Newton's First Law states that "A body remains in its state of rest or of uniform motion in a straight line unless acted upon by an external force."

Newton's Second Law

Newton's Second Law states that "The rate of change of momentum of a body is proportional to the impressed force and takes place in the direction of the force."

$$F \propto \frac{d}{dt}(mv)$$

where F is the impressed force and mv is mass $\times$ velocity (= momentum).

Newton's Third Law

Newton's Third Law states "To every action, there is an equal and opposite reaction."

Parallelogram Law of Forces

The parallelogram law of forces states that "If two forces acting simultaneously at a point are represented in magnitude and direction by the adjacent sides of a parallelogram, then the diagonal of the parallelogram passing through their point of intersection represents the resultant in magnitude and direction."

If P and Q are two forces acting at the point O , then the resultant of P and Q will be R given by

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

Principle of Transmissibility of Forces

The principle of transmissibility of forces states that "The effect of a force on a rigid body will remain unchanged if the point of application of the force is transmitted along its line of action to another point."

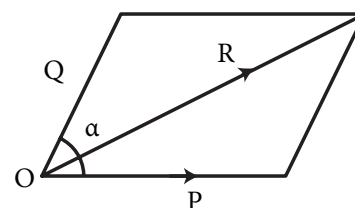
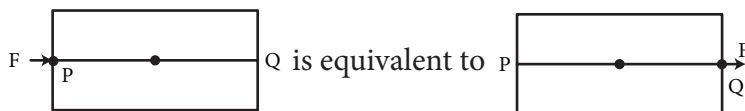
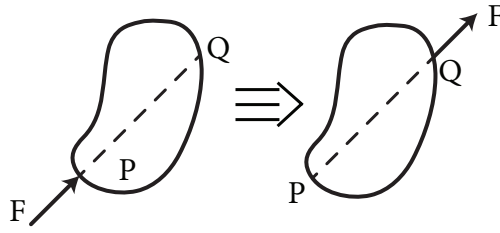


Figure 1.1: Parallelogram of forces



In each of the above cases, the results will not be affected even if the force F is shifted from point P to point Q .

How Do We Resolve Forces?

For analyzing the effect of forces on bodies, it is often convenient to resolve the forces into components that act along any two mutually perpendicular directions.

Suppose a force P acts at an angle of θ with respect to the horizontal as shown in fig.(i).

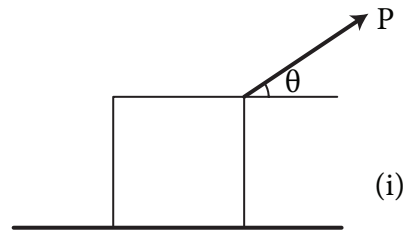
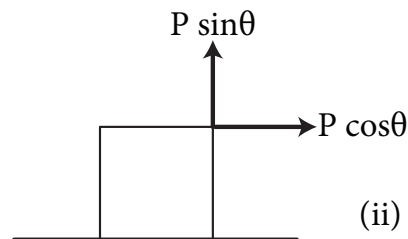


Figure 1.2: Transmissibility of forces

P can be resolved into vertical and horizontal components as shown in fig.(ii). So the figures (i) and (ii) represent equivalent force systems.



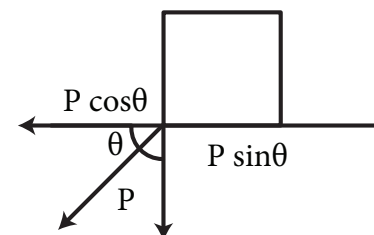
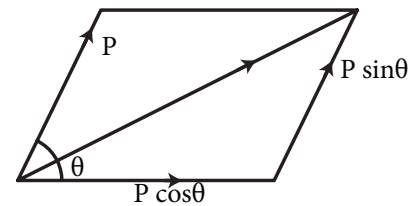
Naturally the resultant of these two components will, in turn, be P .

$$i.e. \sqrt{P^2 \cos^2 \theta + P^2 \sin^2 \theta} = \sqrt{P^2 (1)} = P$$

using Pythagorus theorem.

Here, the force P was upwards and towards the right. So, one component will be upwards and the other will be towards the right. The cosine component will be the one with which P makes an angle θ . The other will be the sine component.

If the force P is downwards and towards the left, one component will be downwards and the other towards the left.



Solved Problems

Problem 1

Find the resultant of the forces acting on the body shown in the figure.

Solution

Taking the forces towards right ($\rightarrow$) as +ve and those towards left ($\leftarrow$) as -ve, the resultant is given by

$$\sum F = 10 + 20 + 5 - 12 = 23 \text{ N} (\rightarrow)$$

Thus, the resultant force has a magnitude of 32 N and acts towards the right.

Problem 2

Find the resultant of the two forces acting at the point P.

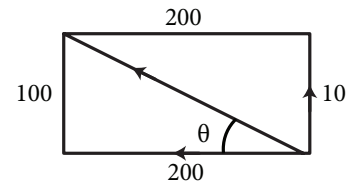
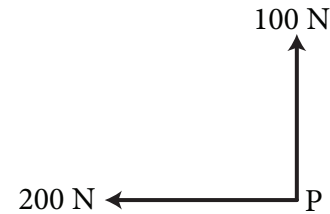
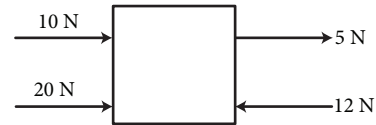
Solution

Using Pythagoras theorem,

$$R = \sqrt{(200)^2 + (100)^2} = 223.6 \text{ N}$$

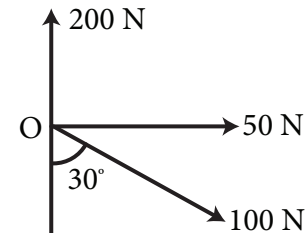
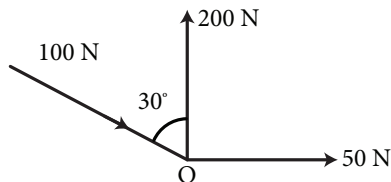
in a direction as shown, where,

$$\theta = \tan^{-1} \frac{100}{200} = 26.6^\circ$$



Problem 3

Find the resultant of the forces acting at the point O.



Solution

200 N and 50 N are pulling forces. So the 100 N force can be represented conveniently as a pulling force.

Resolve 100 N into two mutually perpendicular components – $100 \sin 30$ ($\rightarrow$) and $100 \cos 30$ ($\downarrow$).

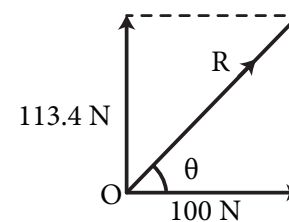
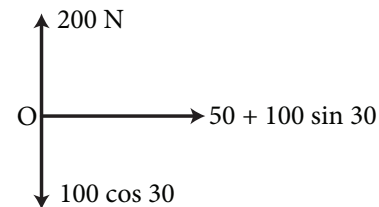
Resultant vertical force

$$\sum F_v = 200 - 100 \cos 30 = 113.4 \text{ N} (\uparrow)$$

Resultant horizontal force

$$\sum F_H = 50 + 100 \sin 30 = 100 \text{ N} (\rightarrow)$$

Now we have only two forces

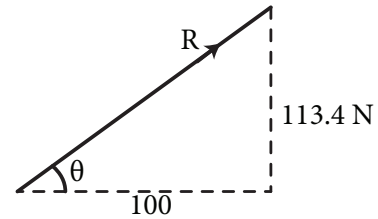


Resultant of these two forces

$$R = \sqrt{(113.4)^2 + (100)^2} = 151.2 \text{ N}$$

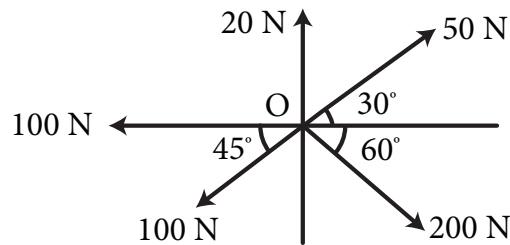
at an angle θ with the horizontal, where θ is given by

$$\tan^{-1} \frac{113.4}{100} = 48.6^\circ$$

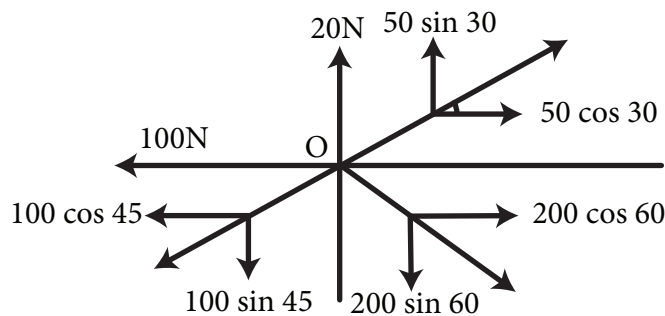


Problem 4

Find the resultant of all the forces acting at the point O.



Resolve all the forces into the vertical and horizontal components.



All these horizontal and vertical forces act at point P

Summing up vertical forces after taking upward forces as +ve and down forces as -ve,

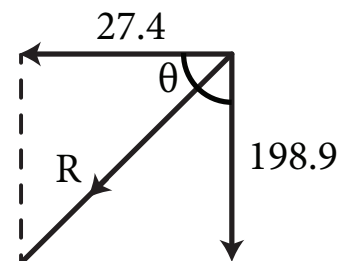
$$\begin{aligned} \sum V &= (20 + 50 \sin 30) - (100 \sin 45 + 200 \sin 60) \\ &= -198.9 \text{ N} = 198.9 (\downarrow) \end{aligned}$$

-ve sign indicates that the force acts in the downward direction. Similarly, taking forces to the right as +ve and a forces to the left as -ve,

$$\begin{aligned} &= (50 \cos 30 + 200 \cos 60) - (100 + 100 \cos 45) \\ &= -27.4 \text{ N} = 27.4 \text{ N} (\leftarrow) \end{aligned}$$

We have two forces, for which the resultant is computed using Pythagorus theorem.

$$R = \sqrt{(27.4)^2 + (198.9)^2} = 200.8 \text{ N}$$

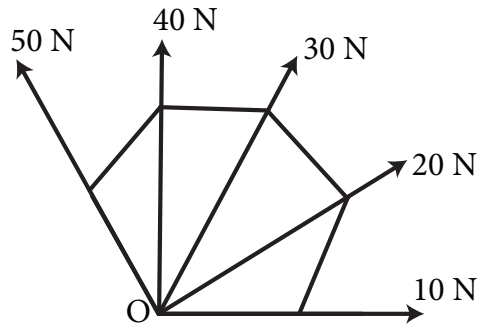


at an angle θ with horizontal when

$$\theta = \tan^{-1} \frac{198.9}{27.4} = 82.2^\circ$$

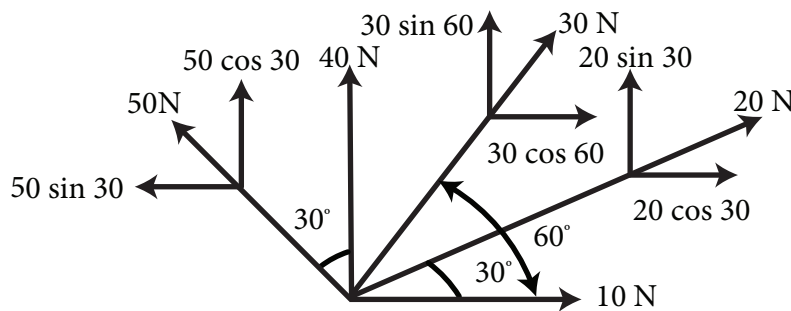
Problem 5

Five forces act on the vertices of a regular hexagon is shown in the figure. Find out the resultant force



Solution

Resolving the forces vertically and horizontally



Taking rightward forces as +ve,

$$\sum H = 20 \cos 30 + 30 \cos 60 - 50 \sin 30 = 7.32 (\rightarrow)$$

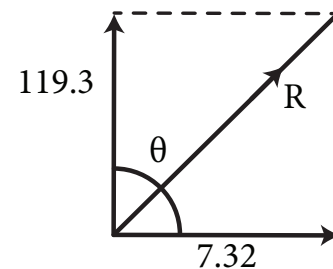
Taking upward forces as +ve,

$$\sum V = 50 \cos 30 + 40 + 30 \sin 60 + 20 \sin 30 = 119.3 \text{ N } (\uparrow)$$

Now, we have two forces for which the resultant can be found as

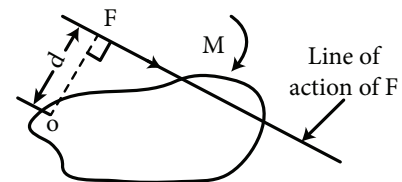
$$R = \sqrt{(7.32)^2 + (119.3)^2} = 119.5 \text{ N}$$

$$\theta = \tan^{-1} \frac{7.32}{119.5} = 3.5^\circ$$



Moment of a force

Moment of a force about a point is the product of a force and the perpendicular distance between the point and the line of action of the force.



If the object given in the figure is assumed to be hinged at O, the effect of the force F obviously is to turn the body in the clockwise direction (↻) about O. This turning effect is quantified using the moment M

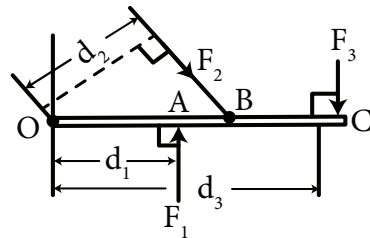
$$M = F \times d$$

where d is the perpendicular distance between O and the line of action of F .

Resultant moment

To find out the resultant moment of a body with respect to a point, we can sum the moments acting on the body algebraically by taking moment in any one direction as positive and the other as negative.

Consider a beam hinged / fixed at the point O as shown. Three forces F_1 , F_2 and F_3 act on it at points A, B, and C. The perpendicular distances from O to the line of action of these forces are d_1 , d_2 and d_3 as shown.



If the beam is assumed to be hinged at O, the effect of the moment produced by F_1 is to rotate the beam in the counter clockwise direction (↺).

Now consider F_2 . The effect of the moment due to F_2 is to rotate the beam in the clockwise (↻) direction about O.

Consider F_3 . Again the effect is to turn the beam in the clockwise direction (↻).

If we take the counter clockwise moments as +ve and the clockwise ones as -ve, the resultant moment M can be found as

$$M = (F_1 \times d_1) - (F_2 \times d_2) - (F_3 \times d_3)$$

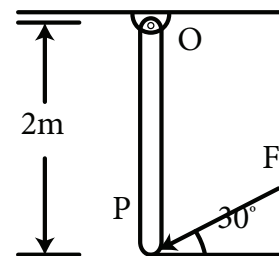
Solved Problems

Problem 1

Find out the moment produced by the force $F = 100$ N about the hinge on the suspended rod shown in the figure.

Solution

Moment of a force about a point is the product of the force and the perpendicular distance between the point and the line of action of the force.



Here, the perpendicular distance can be found as given below.
From the triangle OPQ,

$$OQ = OP \sin 60 = 2 \times (0.866) = 1.732 \text{ m}$$

Moment

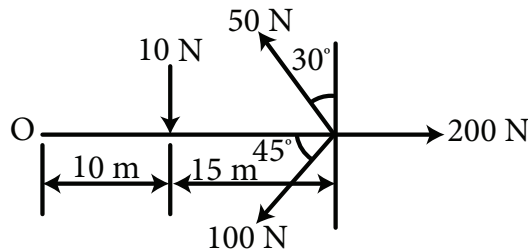
$$M = F \times d = F \times OQ = 100 \times 1.732 = 173.2 \text{ Nm}$$

The effect of the moment is to rotate the rod in a clockwise direction about O.

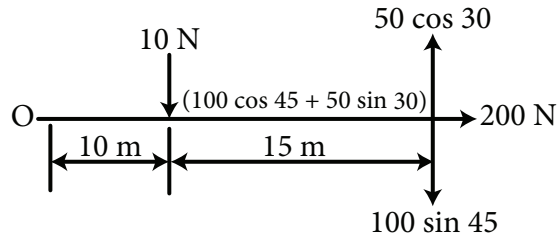
So, it is a clockwise moment.

Problem 2

Referring to the figure, find out the resultant moment at the point O caused by all the forces.



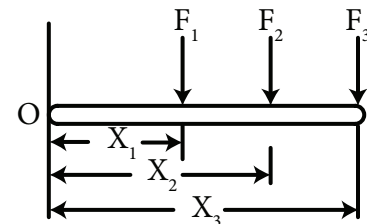
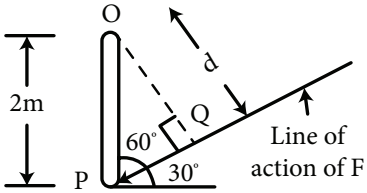
Resolving the inclined forces,



Moment about O is the product of the force and the perpendicular distance between O and the line of action of the force.

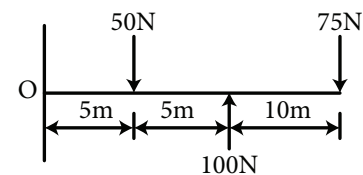
The $100 \cos 45$, $50 \sin 30$ and 200 N forces have their lines of action through O itself.

$$\begin{aligned} \sum M &= (50 \cos 30 \times 25) - (100 \sin 45 \times 25) \\ &\quad - (10 \times 10) + (100 \cos 45 \times 0) \\ &\quad + (50 \sin 30 \times 0) + (200 \times 0) \\ &= -785.2 \text{ Nm} \\ &= 785.2 \text{ Nm} \quad \text{clockwise} (\curvearrowright) \\ &(\because \text{we have taken clockwise as -ve}) \end{aligned}$$



Principle of Moments (Varignon's Theorem)

Varignon's Theorem of Moments states that the moment produced by the resultant of a system of forces acting on a body about a point



will be equal to the algebraic sum of the moments produced by individual forces about the same point.

For example, in the given system of the forces,
About point O

$$M = F_1x_1 + F_2x_2 + F_3x_3$$

If the resultant of F_1 , F_2 and F_3 can be represented by F and its distance is 'x' from O, then,

$$\sum M = Fx \text{ or } F_1x_1 + F_2x_2 + F_3x_3 = Fx$$

Solved Problems

Problem 1

Find the position and magnitude of the resultant of the system of forces given below

Solution

Taking downward forces as +ve,

$$\text{Resultant } R = 50 + 75 - 100 = 25 \text{ N } (\downarrow)$$

To find the position of R, take moments about O,

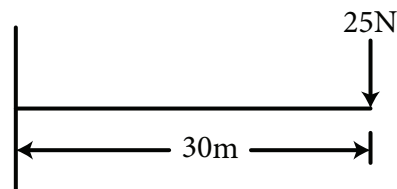
$$\sum M = (100 \times 10) - (50 \times 5) - (75 \times 20) = -750 \text{ Nm}$$

$$\text{is } \sum M = 750 \text{ Nm (clockwise)}$$

But $\sum M = R \times x$ according to Varignon's Theorem.

$$\text{So, } x = \frac{\sum M}{R} = \frac{750}{25} = 30 \text{ m}$$

Therefore the position of R can be represented as



R is downward since $\sum M$ is clockwise.

